МЕЖДУНАРОДНЫЙ УНИВЕРСИТЕТ INTERNATIONAL UNIVERSITY

ХАЛЫҚАРАЛЫҚ УНИВЕРСИТЕТІ

Faculty of Computer Technology and Cyber Security

Laboratory work №3

By discipline: “Fundamentals of Logic Design”

On the topic: “First customized 3D model”

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Introduction

The purpose of this laboratory work is to create a customized 3D model based on what we have learned in the previous work. My choice is a chess piece as it represents all the knowledge I have acquired.

Workflow

My model lay down on Front plane. I created a half of two-dimensional model of chess figure.

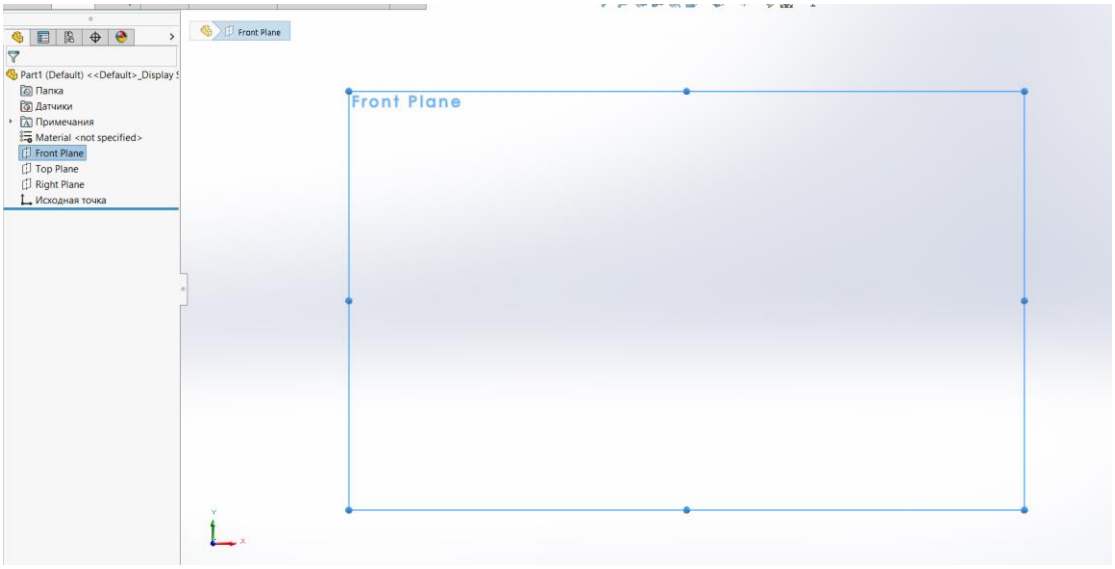


Figure 1.1 – Front Plane

My model consist of eight parts. The dimensions of the figure's foundation are 30mm long and 10mm wide, the top of 26.45mm and 5mm long and wide respectively.

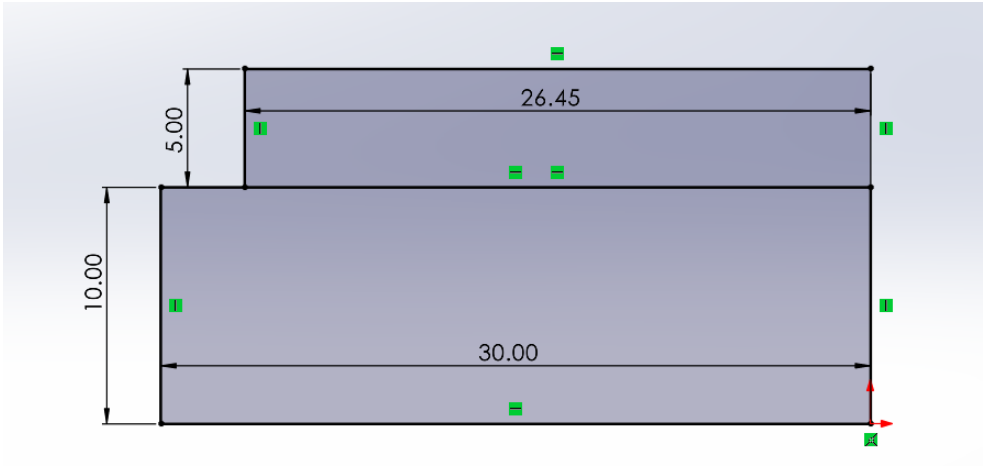


Figure 1.2 – Figure's Foundation

The next step is to draw the Centreline 73mm long.

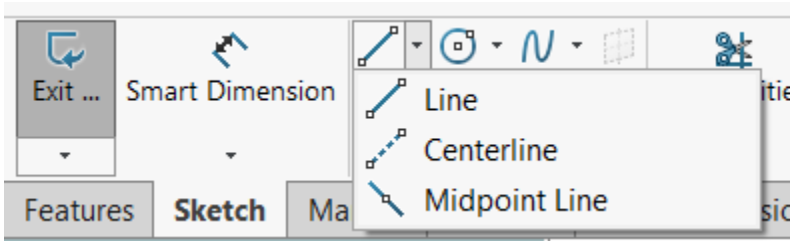


Figure 1.3 – Centerline tool

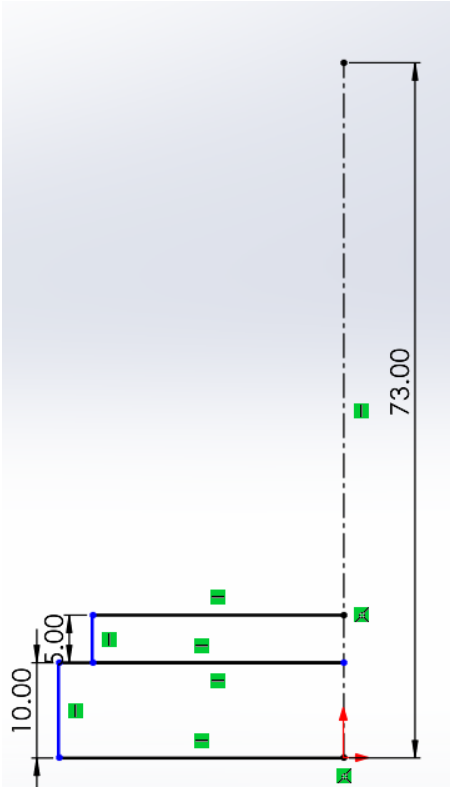


Figure 1.4 – Centerline of sketch

To create the figure's hat, two rectangles are created. The parameters of the lower one are 18mm in length and 3mm in width. The second one has 15mm and 3.25mm in length and width respectively. The distance from the foundation to the hat is 35mm.

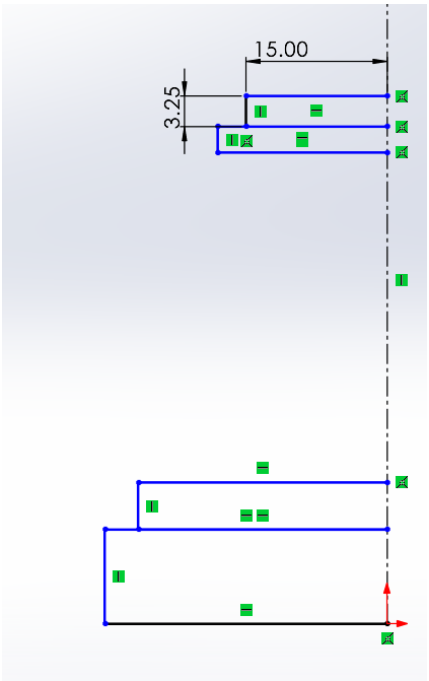


Figure 1.5 – Figure's hat

To get rounded sides and a hat for the chess piece, I used the Spline tool. However, to make a circle, you need to use the Circle tool.

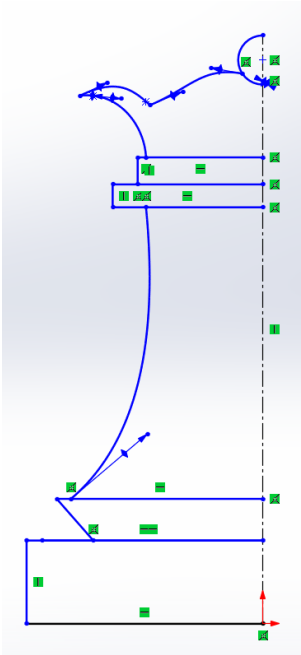


Figure 1.6 – Figure

Trim Entities is used to remove extra lines to leave only the side.

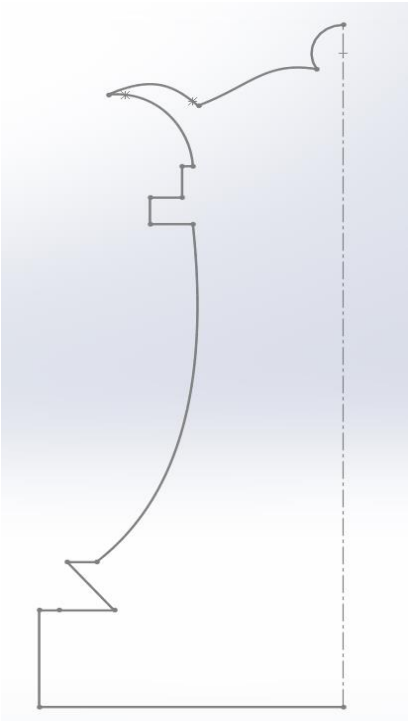


Figure 1.6 – Figure

The next tool is Revolved Base, which rotates selected sketch outlines around an axis to create a solid pattern. In this sketch, Centreline is selected as the axis.

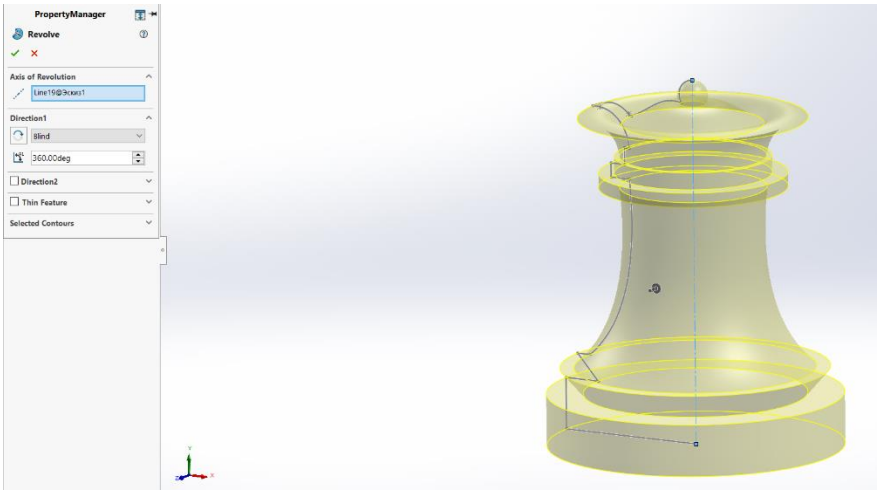


Figure 1.7 – Revolved Base

The figure is almost ready. It remains to round the edges of the foundation and the bottom part of the hat.

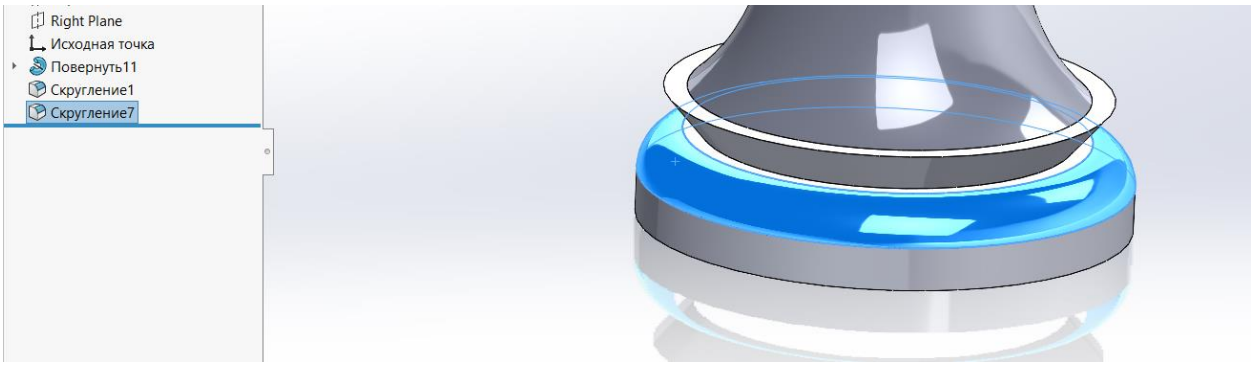


Figure 1.8 – The rounded foundation

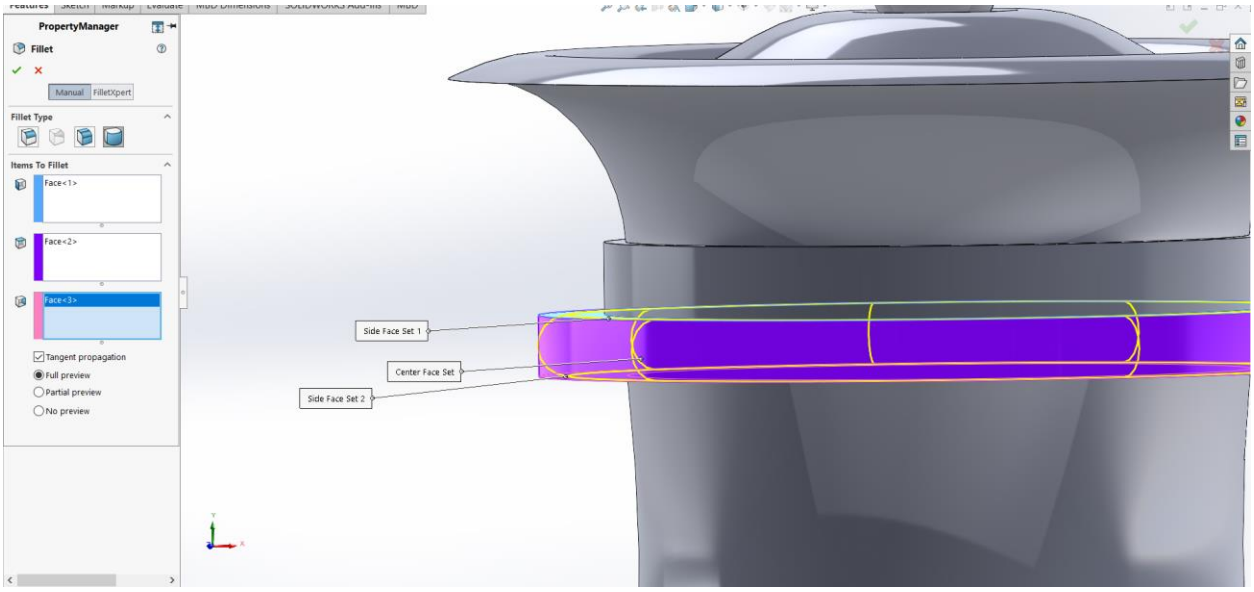


Figure 1.8 – The Rounded base of hat



Figure 1.8 – Finished model of a chess piece

Conclusion

I created a simple chess figure that took me three tries. I chose to create a figure based on half of a two-dimensional piece, since the first attempt failed. My first chess piece was created using only three-dimensional rectangles and squares, and as a result, it was not neat. The second and third figures were created in the same way, which made them more realistic. Creating a model does not take a lot of time if you initially calculate the size of each part and the stage of creation. In my work, I did not do this, from now on I will make sketches of the figure with dimensions.